Ca-STANet: Spatiotemporal Attention Network for Chlorophyll-a Prediction With Gap-Filled Remote Sensing Data

Min Ye, Bohan Li, Jie Nie, *Member, IEEE*, Yuntao Qian, *Senior Member, IEEE*, and Lie-Liang Yang, *Fellow, IEEE*

***Abstract*— Long-term chlorophyll-a (Chl-a) prediction has the potential to provide an early warning of red tide and support fishery management and marine ecosystem health. The existing learning-based Chl-a prediction methods mostly predict a single point or multiple points with monitoring data. However, the monitoring data are subject to sparse sampling and difficult to be measured in a large-scale and synchronous way. Moreover, the advanced learning-based models for point Chl-a prediction, such as long short-term memory (LSTM) and convolutional neural network (CNN)-LSTM, are unable to fully mine the spatiotemporal correlation of Chl-a variations. Therefore, by using the satellite remote sensing data with extensive coverage, we design a framework, namely, Ca-STANet, to simultaneously predict the Chl-a of all the locations in a large-scale area from the perspective of spatiotemporal field. Specifically, in our method, the original data are first divided into multiple subregions to capture the spatial heterogeneity of large-scale area. Then, two modules are, respectively, operated to mine the spatial correlation and long-term dependency features. Finally, the outputs from the two modules are integrated by a fusion module to fully mine the spatiotemporal correlations, which are exploited to attain the final Chl-a prediction. In this article, the proposed Ca-STANet is comprehensively evaluated and compared with the legacy methods based on the OC-CCI Chl-a 5.0 data of the Bohai Sea. The results demonstrate that the proposed Ca-STANet is highly effective for Chl-a prediction and achieves higher prediction accuracy than the baseline methods. Moreover, as the OC-CCI Chl-a 5.0 data have many missing areas, we introduce DINEOF method to fill the data gaps before using them for prediction.**

***Index Terms*— Chlorophyll-a (Chl-a), convolutional neural network, deep learning (DL), remote sensing data, spatiotemporal attention, spatiotemporal prediction.**

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Min Ye and Jie Nie are with the Department of Computer Science and Technology, Ocean University of China, Qingdao 266100, China (e-mail: yemin@stu.ouc.edu.cn; niejie@ouc.edu.cn).

Bohan Li and Lie-Liang Yang are with the School of Electronics and Computer Science, University of Southampton, SO17 1BJ Southampton, U.K. (e-mail: bl2n18@ecs.soton.ac.uk; lly@ecs.soton.ac.uk).

Yuntao Qian is with the College of Computer Science, Zhejiang University, Hangzhou 310027, China (e-mail: ytqian@zju.edu.cn).

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I. Introduction

In THE ocean, the level of chlorophyll-a (Chl-a) has been demonstrated to be a key indicator of ecosystem changes, since it is used as a proxy for phytoplankton biomass [1], and also, a paramount variable for studying the environmental effects arose from the ocean dynamic process [2]. Moreover, Chl-a has a significant impact on the stability of the ocean ecosystem, the exchange of carbon dioxide flux across the air–sea interface, and the distribution of marine aquatic resources [3], [4], [5]. Therefore, long-term and reliable Chl-a prediction is highly important for the research of global carbon cycle and the utilization of fishery resources, as well as for the timely warning of red tide disasters [6].

In general, Chl-a prediction can be deemed as a problem of multivariate time-series forecasting, which exploits the historical Chl-a data and exogenous factors (e.g., temperature and PH) to predict the future Chl-a. To date, the classical approaches on Chl-a prediction can be roughly classified into two categories, namely, the time-series analysis method and the physics-based method. Specifically, Chen et al. [7] investigated the algal bloom prediction based on an autoregressive integrated moving average (ARIMA) model. However, the spatial–temporal evolution of Chl-a is, in general, nonlinear due to various factors, such as sea surface temperature, wind speed, and the light transmittance of seawater [8], [9]. As the existing time-series analysis methods are designed mainly for extracting inherent characteristics via linear evolution, they are incapable of providing the accurate prediction of Chl-a.

On the other side, the physics-based methods conduct Chl-a prediction mainly based on the ecological dynamics and flow-diffusion equation [10]. For instance, Park et al. [11] introduced the Earth system model to predict the seasonal and multiyear ocean chlorophyll. The National Aeronautics and Space Administration (NASA) established a global biogeochemical prediction model, which was used to predict the global ocean chlorophyll over a nine-month period [12]. To date, the physics-based methods have been comprehensively developed with the motivation to achieve high resolution and simultaneously deal with multiphysical processes. However, we should mention that these physics-based methods usually demand accurate knowledge about a large number of variables as well as their complicated relationships, making it hard to build a comprehensive model. Furthermore, the errors

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accumulated during prediction may hinder the model from reliable long-term prediction in practical implementation.

Recently, the learning-based methods for Chl-a prediction have captured a lot of research attention, as the result that the techniques for Chl-a data acquisition have been well developed, and hence, a lot of data are available. Initially, the learning-based methods introduced for Chl-a prediction include, e.g., random forest, support vector machine, and artificial neural network (ANN) [13], [14], [15], [16], [17]. These methods usually have a relatively small number of hidden layers and can be classified as the shallow learning-based methods (SLBM) [18]. Furthermore, the ensemble learning-based models have been designed for Chl-a prediction. For instance, in [19], different wavelet ANNs were integrated with the aid of the least square boosting ensemble and Bates–Granger techniques, which are capable of achieving more reliable Chl-a prediction than individual ANN. However, the abovementioned methods are, in general, only feasible for handling the low-dimensional data with simple nonlinear associations. They are not efficient for representing the dynamic evolution of Chl-a [20].

By contrast, the deep learning (DL) methods, which use multiple processing layers to learn the representations of data via multiple levels of abstraction, can exploit the extremely intricate functions of inputs [21]. Thus, they outperform the SLBM and have become the mainstream approaches for Chl-a prediction. For example, in [22], the recurrent neural network (RNN) was introduced for predicting the Chl-a in the Nakdong River and demonstrated to provide a higher prediction accuracy than the SLBM. The long short-term memory (LSTM) model was introduced to predict Chl-a in [23], [24], [25], and [26]. As an enhanced RNN model, the LSTM with the long-term memory obtained by several control gates is feasible for solving the sequence modeling problems. To be a little more specific, Cho et al. [23] used the LSTM for the multistep Chl-a prediction in Gongju, South Korea. The studies suggested that the LSTM is a high-efficiency model, which can exploit the temporal Chl-a variation. To improve the accuracy of prediction, Barzegar et al. [27] performed the Chl-a prediction at the Prespa lake using the hybrid convolutional neural network (CNN)-LSTM model. The studies revealed that the hybrid CNN-LSTM model outperforms the individual CNN or LSTM models in terms of the prediction accuracy.

The aforementioned learning-based methods mainly predict Chl-a at a single point or multiple points, as shown in Fig. 1. They are not suitable for the large-scale area Chl-a prediction, due to the fact that only a small number of points are involved in a model, and each point is individually fed to the model for training. Thus, the spatiotemporal correlations within the whole area cannot be fully exploited. To achieve efficient Chl-a prediction in large-scale areas, the spatiotemporal field, as illustrated in Fig. 1, combined with the DL methods is proposed for Chl-a prediction in this article. Explicitly, the large-scale area Chl-a prediction requires continuous spatiotemporal grid data, while the on-site sampling and buoy-based data, which are difficult to be measured in a large-scale and synchronous way, cannot satisfy this requirement. Therefore, in our studies, the satellite remote sensing data,

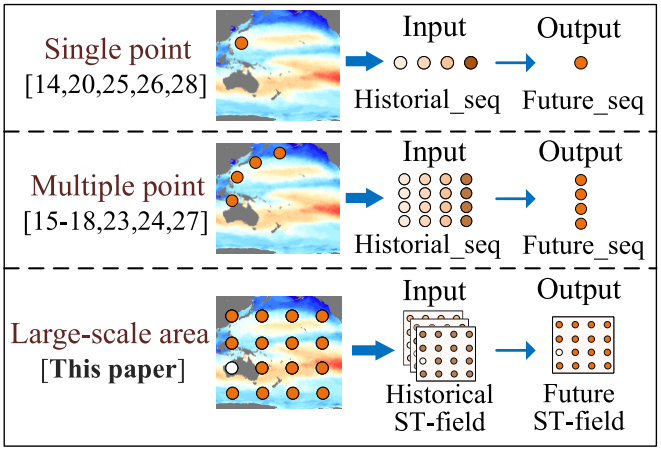


Fig. 1. Typical Chl-a prediction categories classified based on the learning-based methods. Historical_seq means the time-domain historical sequences, Future_seq means the future sequences, and ST field means the spatiotemporal field.

which have the features of extensive coverage and the capability of realizing spatiotemporal continuous observation, will be used to construct the spatiotemporal grid data for Chl-a prediction.

However, in practice, the cloud obscuring measurements and the contamination of high sun glint may result in the loss of some satellite ocean color data of Chl-a. Hence, filling the remote sensing data gaps is indispensable for understanding the hydrodynamics and biophysical interactions of the ocean. In the open literature, various methods have been introduced for recovering missing data, including the spline interpolation [28], optimal interpolation (OI) [29], Kriging interpolation [30], data interpolating empirical orthogonal functions (DINEOF) [31], and the DL methods [32]. As demonstrated in [33], DINEOF can achieve similar filling results as the OI approach, but it is 30 times faster. Henn et al. [34] compared DINEOF with a temporal interpolation method and other three methods based on spatial interpolation. It was observed that DINEOF is the most accurate method for missing data recovery in large-scale areas. Finally, it is shown [32] that the CNN-relied DL approach can achieve better performance than the DINEOF method; however, it depends on a large amount of evenly distributed in situ data for training. In addition, the DINEOF method has been widely applied in reconstructing Chl-a data, such as Moderate-Resolution Imaging Spectroradiometer (MODIS) and Sea-viewing Wide Field-of-view Sensor (SeaWiFS), in the Bohai and Yellow Seas [35], [36], [37]. Therefore, in this article, the DINEOF method is introduced to fill the missing data in the Bohai Sea.

Having the spatiotemporal relied data prepared, the next step is to introduce a predictor, which can efficiently use the data to make a reliable Chl-a prediction in large-scale areas. To this regard, we should note that the abovementioned advanced LSTM and CNN-LSTM models for point-source Chl-a prediction are unable to fully mine the spatiotemporal correlation of the Chl-a variations. By contrast, the DL-assisted spatiotemporal methods have received a lot of research attention for prediction in large-scale areas. Zheng et al. [38] proposed

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a multiscale CNN network for forecasting the sea surface temperature (SST) in the eastern equatorial Pacific Ocean. This kind of methods [38], [39], [40] developed based on CNN can effectively improve the mining ability of spatial correlation within the whole area through the multiple convolution and downsampling operations. Moreover, Zhou et al. [41] proposed a multilayer fusion RNN, which uses a cell to fuse the global and local spatiotemporal features, to predict the sea surface height anomaly (SSHA) areas. It was shown that the RNN-assisted methods [41], [42], [43], such as Convlstm and MLFrnn, are capable of extracting the information of spatiotemporal evolutions from neighboring grids.

However, the abovementioned DL-assisted spatiotemporal prediction methods cannot fully exploit the evolution features in the respective domains of space and time. More recently, Ravuri et al. [44] proposed a deep generative model with the dual spatial and temporal discriminator for precipitation nowcasting, showing the improved forecast quality and forecast consistency. Inspired by the work in [44] and combined with the spatiotemporal characteristics of Chl-a evolution, in this article, we design a framework to simultaneously predict the Chl-a of all the locations in a large-scale area by a spatiotemporal feature fusion network, which is termed the Ca-STANet. In our method, the original data are first divided into multiple subregions to capture the spatial heterogeneity in large-scale area prediction. Then, for each subregion, two modules, namely, the spatial attention feature extraction (SAFE) and temporal attention feature extraction (TAFE), are constructed to enhance the feature extraction of the spatial variation and temporal dependency. Moreover, each of the subregions is trained independently so as to avoid the possible catastrophic forgetting caused by a single model-based training. In our double-module-based training, the SAFE module introduces a spatial attention mechanism based on the subregion's inputs and global inputs, aiming at extracting the features of the spatial variation within a subregion, but at the same time, exploiting the global dynamic correlation between subregions. By contrast, the TAFE module focuses on extracting the features in the time domain to capture the long-term dependency existing in the data. After that, the outputs from the two modules are integrated by a fusion module, where the spatiotemporal correlation is fully mined for the Chl-a prediction. Finally, the predicted results from subregions are combined to provide the Chl-a prediction of the whole large-scale area.

The main contributions of this article are summarized as follows.

1) Different from the existing point source Chl-a prediction, the Chl-a prediction in large-scale areas is investigated with the aid of the DL models operated in spatiotemporal field.

2) The DINEOF method is introduced to fill the five-day and monthly missing Chl-a data obtained by the remote sensing OC-CCI 5.0 in Bohai Sea. The impact of the gap-filled data in winter on the Chl-a prediction is analyzed.

3) A Ca-STANet is proposed to provide the Chl-a prediction in large-scale areas. In our Ca-STANet, an original large-scale area is first divided into multiple subregions to capture the spatial heterogeneity. Then, the SAFE and TAFE modules are constructed to enhance the feature extraction of spatial variation and temporal dependency. Finally, a fusion module is used to mine the spatiotemporal correlation from the outputs provided by the two modules.

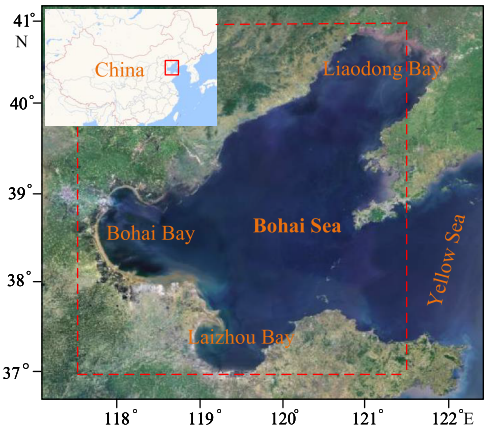


Fig. 2. Area of Bohai Sea studied (in the red rectangle).

4) The effectiveness of the proposed method is validated by comparing our method with the state-of-the-art prediction methods, demonstrating that our method is capable of efficiently mining the spatiotemporal correlations and, hence, providing higher accuracy of Chl-a prediction.

The remainder of this article is organized as follows. Section II presents the study area and datasets. Section III details the scheme for preparing the Chl-a data and the proposed Ca-STANet framework for Chl-a prediction. Section IV reports the experimental results and provides our analysis. Section V further analyzes the prediction results of winter data and their influence on the model. Finally, in Section VI, we summarize the research observations and discuss the possible future research issues.

II. Data

A. Study Area

Bohai Sea is located in the north of China (bounded by 37°–41°N, 117.5°–121.5°E), as shown in Fig. 2. It is a shallow shelf sea, with an average water depth of about 18 *m* and a total area of about 77 000 km^2. Under the influence of human activities in Bohai Rim area, the eutrophication of seawater has become serious, and the phytoplankton biomass has increased rapidly. The distribution of Chl-a shows a pattern of high concentration in the shallow waters near land, and the Chl-a gradually decreases when moving away from the shores [45].

B. Datasets

In this study, the five-day and monthly surface Chl-a 5.0 data of Bohai sea with a spatial resolution of 4 km were downloaded from the Ocean Color Climate Change Initiative (OC-CCI), which is available at http://www.esa-oceancolour-cci.org. The Chl-a 5.0 dataset is obtained by merging the data from Medium-Resolution Imaging Spectrometer (MERIS), MODIS, SeaWiFS, and Visible Infrared Imaging Radiometer

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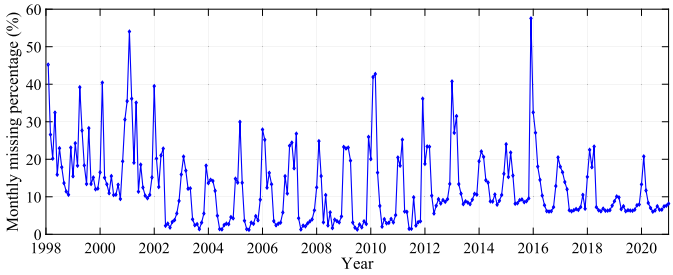


Fig. 3. Time series showing the area-averaged missing rate of the monthly Chl-a data for the Bohai Sea.

Suite (VIIRS) data. In this article, the Chl-a dataset covers the period from 1998 to 2020.

III. Methods

A. Data Preprocessing and Gap Filling

The loss of Chl-a data due to clouds and sun glint is different in different time periods. It is found that during the 23 years considered, the mean missing rate of five-day and monthly Chl-a data is 45.8% and 12.6%, respectively. For example, Fig. 3 shows the missing percentage of monthly Chl-a data in the Bohai Sea, which is uneven in both time and space. In particular, sea ice and heavy clouds contribute to the high loss rate of Chl-a data in winter. In order to avoid the loss of important local information and achieve more accurate prediction, the missing data should be appropriately reconstructed.

Therefore, the DINEOF method [31] is introduced to first reconstruct the missing data. In principle, DINEOF uses an empirical orthogonal function (EOF) for reconstructing the missing data in a geophysical dataset, via deciphering the dominant variability modes within the data [46]. The advantages of this method include that the following hold: 1) it does not require the priori information about the decorrelation scale and 2) it can exploit the multiple data types with inherent correlation to increase data coverage.

The DINEOF method can be described by the following four steps.

1) Initial Data Processing: Consider an initial matrix $\bar{\boldsymbol{X}}$, where the rows denote the number of spatial points and the columns denote the length of time. A data matrix $\boldsymbol{X}$ is then obtained by subtracting the mean value from $\bar{\boldsymbol{X}}$, while setting the initially missing data to zero.

2) Iterative Data Replacement: An EOF decomposition is performed on $\boldsymbol{X}$, based on which the values in $\boldsymbol{X}$ are updated using the following equation:

$$\boldsymbol{X}_{i,j} = \sum_{k=1}^{K} \rho_k (\boldsymbol{u}_k)_i \left(\boldsymbol{v}_k^T\right)_j \tag{1}$$

where i and j are the spatial and temporal indexes of the missing data in $\boldsymbol{X}$, K represents the number of EOF modes, $\boldsymbol{u}_k$ and $\boldsymbol{v}_k^T$ are the spatial and temporal functions of the corresponding EOF mode, and ρ_k is the corresponding singular value. The above EOF decomposition and data updating of $\boldsymbol{X}$ are repeated until convergence is achieved. In our study, the convergence is assumed to be achieved, when the maximum error between the input $\boldsymbol{X}_{i,j}$ values and their reconstruction ones reaches a Lanczos convergence threshold of 10^{-8}.

3) Finding the Optimal Number of EOF Modes: Step 2) is repeated with $K = 1, 2, \ldots, K_{\max}$, and the correspondingly reconstructed matrices $\boldsymbol{X}_K$ are obtained. Then, the optimal number of EOF modes (denoted as K_{opt}) is obtained through a cross-validation technique [47]. Specifically, K_{opt} is chosen as the K value that minimizes the error between the dataset left for validation and their correspondingly reconstructed values.

4) Gap Filling of Missing Data: Once K_{opt} is obtained, Steps 1) and 2) are operated again on the basis of K_{opt} and $\bar{\boldsymbol{X}}$ to obtain a $\boldsymbol{X}$, where $\bar{\boldsymbol{X}}$ also includes the data previously set aside for cross validation. Finally, the mean value of $\bar{\boldsymbol{X}}$ is added back to $\boldsymbol{X}$ to obtain the finally reconstructed gap-free data.

B. Formulation of the Chl-a Prediction Problem

We assume a spatial region represented by an $M \times N$ grid, which consists of M rows and N columns. Let $\boldsymbol{S}_{1:T} = (\boldsymbol{I}_1, \boldsymbol{I}_2, \ldots, \boldsymbol{I}_T)$ be the dynamic Chl-a sequence of length T, where $\boldsymbol{I}_t \in \mathbb{R}^{M \times N}$ contains the Chl-a values of the tth frame. Hence, $\boldsymbol{S}_{1:T}$ is a 3-D vector of size (T, M, and N). The goal of Chl-a prediction is to predict a future Chl-a sequence up to L steps subsequent to the historical satellite remote sensing sequence $\boldsymbol{S}_{1:T}$, which can be formulated as follows:

$$\widehat{\boldsymbol{S}}_{T+1:T+L} = \arg\max_{\boldsymbol{S}_{T+1:T+L}} p(\boldsymbol{S}_{T+1:T+L} \mid \boldsymbol{S}_{1:T}) \tag{2}$$

where $\widehat{\boldsymbol{S}}_{T+1:T+L} = (\hat{\boldsymbol{I}}_{T+1}, \ldots, \hat{\boldsymbol{I}}_{T+L})$ represents the predicted Chl-a sequence of length L.

C. Structure of Prediction Model

This section describes in detail the proposed Ca-STANet, which realizes reliable long-term Chl-a prediction by making efficient use of the spatiotemporal correlation features existing in the satellite sensed Chl-a data. The overall structure of Ca-STANet is shown in Fig. 4(a). The inputs to the Ca-STANet are the historical data $\boldsymbol{S}_{1:T}$, as previously defined. As shown in Fig. 4(a), $\boldsymbol{S}_{1:T}$ is first divided into subregions to tackle the spatial heterogeneity in the spatiotemporal input preprocessing (STIP) module. Then, for each subregion, the SAFE module and the TAFE module are, respectively, operated to extract the features of spatial variation and temporal dependency. After that, a fusion module is used to mine the spatiotemporal correlations for the Chl-a prediction, which is referred as the spatiotemporal fusion prediction (STFP) module. Finally, the results from multiple subregions are concatenated to provide the Chl-a prediction of the whole region considered. Below, we analyze these modules one-by-one in detail.

1) STIP Module: The objective of this module is to subdivide the original area and generate the required inputs for the TAFE and SAFE modules. From the Chl-a data of Bohai Sea, we noticed that the spatial relationship within each subregion may vary greatly. As illustrated in Fig. 5, the point Chl-a values in subregions B and D of the offshore water far away from land are similar and have a low level of fluctuation. They show a high degree of positive correlation between nearby points and even between the distant points. By contrast,

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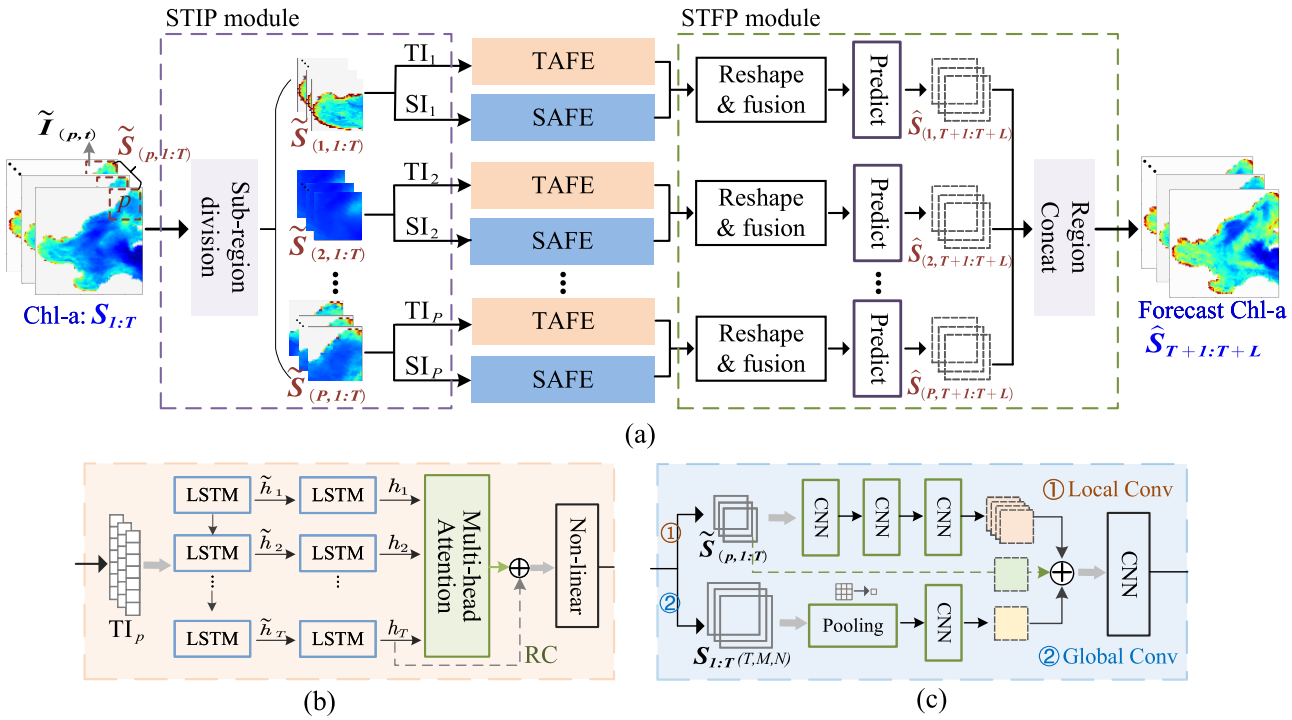


Fig. 4. Overall network structure of Ca-STANet, including mainly the STIP module, TAFE module, SAFE module, and STFP module. (a) Flow chart of Chl-a prediction. (b) TAFE module. (c) SAFE module.

in subregions A and C located at the shallow waters near land, the point Chl-a values are usually different and vary significantly in different seasons. These different correlation relationships existing in the various subregions imply the phenomenon of spatial heterogeneity, which is defined as the uneven distribution of a trait, event, or relationship over a region [48]. However, the spatial heterogeneity of subregions may be overlooked, if the data of a whole region are used as the inputs. To solve this problem, in our method, each of the subregions is trained independently, so as to capture the detailed spatial heterogeneity, while also avoiding the occurrence of the possible catastrophic forgetting caused by using a single model-based training for all subregions.

In detail, each $\boldsymbol{I}_t$ of the original Chl-a inputs, which provides the data of a whole area, is divided into P subregions. Considering the rectangular type of inputs of CNN layers in the SAFE module and the computational efficiency, the area of Bohai Sea is evenly segmented into P subregions. The data of these subregions are denoted as $\widetilde{\boldsymbol{I}}_{(p,t)}$ with $p = 1, \ldots, P$, and we have $\boldsymbol{I}_t = \cup_{p=1}^{P} \widetilde{\boldsymbol{I}}_{(p,t)}$. For one subregion p, the historical sequences are now expressed as $\widetilde{\boldsymbol{S}}_{(p,1:T)} = (\widetilde{\boldsymbol{I}}_{(p,1)}, \widetilde{\boldsymbol{I}}_{(p,2)}, \ldots, \widetilde{\boldsymbol{I}}_{(p,T)})$, as shown in leftmost box of Fig. 4(a). The historical sequence $\widetilde{\boldsymbol{S}}_{(p,1:T)}$ and the global historical sequence $\boldsymbol{S}_{1:T}$ are encapsulated as the inputs to the SAFE module, which are expressed as SI_p in Fig. 4(a). On the other side, an $\widetilde{\boldsymbol{S}}_{(p,1:T)}$ is converted to the 2-D vector of size $(T, (M \times N)/P)$, which produces the inputs to the TAFE module, shown as TI_p in Fig. 4(a). Let us now detail the operations of the TAFE and SAFE modules.

2) TAFE Module: The purpose of the TAFE module is mainly to capture the long-term dependency of subregions and generate temporal features for the following STFP module.

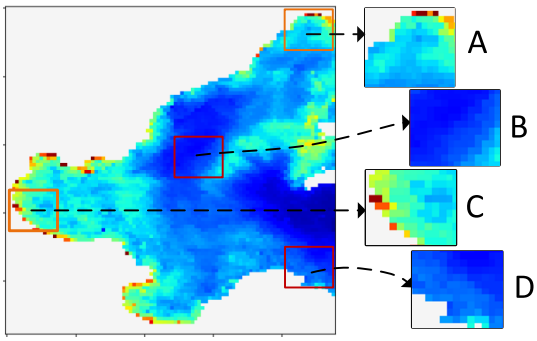


Fig. 5. Example of different levels of spatial heterogeneity throughout the containing subregions.

As shown in Fig. 4(b), the TAFE module consists of the LSTM layers, a multihead attention layer, and a "nonlinear" block. It is well recognized that the LSTM architecture is feasible for capturing the long-term dependency in a sequential pattern. Hence, in this study, we use LSTM as a building block for the TAFE module. Specifically, in the TAFE module, the hidden vectors $\boldsymbol{H}_T = \{\boldsymbol{h}_1, \boldsymbol{h}_2, \ldots, \boldsymbol{h}_T\}$ are first obtained with the aid of a two-layer LSTM network, as shown in Fig. 4(b). Then, the multihead attention layer is executed to learn the internal characteristics and time-dependence between different $\boldsymbol{h}_t$ values. After this learning, the weights for different $\boldsymbol{h}_t$ are reassigned to make the network focus on the relatively important time features. This is achieved via the residual connection (RC) [49] and the output of the multihead attention layer shown in Fig. 4(b). Finally, we express the output of the TAFE module as $\boldsymbol{T}_{\text{tem}}$, which is the output of the "nonlinear" block. Note that, the "nonlinear" block includes the fully connected layers associated with the rectified linear unit (ReLU) function [50].

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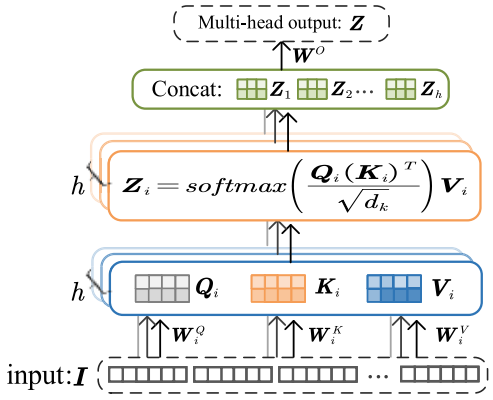


Fig. 6. Structure of multihead attention mechanism.

More details related to the multihead attention layer, defined as an ensemble of h different self-attention blocks, can be found in Fig. 6. Specifically, the input matrix $\boldsymbol{I}$ is first mapped to the h different subspaces through linear transformation as follows [51]:

$$\begin{aligned}[\boldsymbol{Q}_1,\ldots,\boldsymbol{Q}_h]&=\left[\boldsymbol{I}\boldsymbol{W}_1^Q,\ldots,\boldsymbol{I}\boldsymbol{W}_h^Q\right]\\ [\boldsymbol{K}_1,\ldots,\boldsymbol{K}_h]&=\left[\boldsymbol{I}\boldsymbol{W}_1^K,\ldots,\boldsymbol{I}\boldsymbol{W}_h^K\right]\\ [\boldsymbol{V}_1,\ldots,\boldsymbol{V}_h]&=\left[\boldsymbol{I}\boldsymbol{W}_1^V,\ldots,\boldsymbol{I}\boldsymbol{W}_h^V\right]\end{aligned}\tag{3}$$

where $\boldsymbol{Q}_i$, $\boldsymbol{K}_i$, and $\boldsymbol{V}_i$, $i\in[1,h]$, are the query, key, and value matrices of each subspace, and $\boldsymbol{W}_i^Q$, $\boldsymbol{W}_i^K$, and $\boldsymbol{W}_i^V$, $i\in[1,h]$, are the learnable conversion matrices.

Then, as shown in Fig. 6, the attention value of each subspace is computed as follows:

$$\boldsymbol{Z}_i=\text{softmax}\left(\frac{\boldsymbol{Q}_i(\boldsymbol{K}_i)^T}{\sqrt{d_k}}\right)\boldsymbol{V}_i\tag{4}$$

where $\boldsymbol{Z}_i$ is the attention matrix of the ith subspace and $(d_k)^{1/2}$ is applied to change the attention matrix to follow the standard normal distribution so as to achieve the gradient stability. As shown in Fig. 6, $\boldsymbol{Z}_i$ is obtained by multiplying $\boldsymbol{V}_i$ and the attention coefficient that is generated by the softmax. The inputs $\boldsymbol{I}$ are given by the hidden vectors in $\boldsymbol{H}_T$, as previously defined. In this way, the short- and long-term memory features generated by LSTM are further processed in the multihead attention layer.

Finally, the attention matrices $\boldsymbol{Z}_i$ are concatenated and projected to produce the multihead output as follows:

$$\boldsymbol{Z}=\text{Concat}(\boldsymbol{Z}_1,\ldots,\boldsymbol{Z}_h)\boldsymbol{W}^O\tag{5}$$

where $\boldsymbol{W}^O$ is the learnable weight matrix and Concat$(\cdot)$ is the concatenation operation.

Note that in our work, we set $h=2$ and $(d_k)^{1/2}=32$.

3) SAFE Module: The function of this module is mainly to capture the spatial correlation existing in a subregion and between subregions and to generate spatial features for the following STFP module. Our SAFE module is designed to include both the "Local Conv" and "Global Conv" as noted in Fig. 4(c). The "Local Conv" pays attention on extracting the features of the spatial variation within a subregion, while the "Global Conv" emphasizes on deriving the global spatial correlation between subregions. Furthermore, as the inputs to the SAFE module represent the historically evolving data, the SAFE module can also mine the spatiotemporal correlation of the input data of different subregions.

To be more specific, as shown in Fig. 4(c), the values of $\widetilde{\boldsymbol{S}}_{(p,1:T)}$ of SI_p, as previously defined in the STIP module as seen in Fig. 4(a), are input to the "Local Conv" to extract the spatial correlation features within the subregions. The "Local Conv" consists of three cascaded CNN layers, each of which is followed by a ReLU function. The convolution kernel used in the first CNN layer has the size of 5×5, while that used in the following two CNN layers has the size of 3×3. The numbers of convolution kernels of these three cascaded CNN layers are 20, 20, and 30, respectively. By contrast, as the "Global Conv" submodule is introduced to extract the Chl-a variation between different subregions, the data input to the submodule is only the $\boldsymbol{S}_{1:T}$. The functions of this submodule are achieved by one downsampling layer and one CNN layer. The downsampling layer uses the AvgPool to compress information, and the convolution kernel size of the CNN layer is 3×3. In addition, considering that the input Chl-a data at the present time step have a higher correlation with the future Chl-a data, the input Chl-a data at the present time step, i.e., $\widetilde{\boldsymbol{I}}_{(p,T)}$ in $\widetilde{\boldsymbol{S}}_{(p,1:T)}$, shown as the green feature map in Fig. 4(c), are directly linked to the last CNN layer along with the features obtained from both the "Local Conv" [shown by the orange feature map in Fig. 4(c)] and the "Global Conv" [shown by the yellow feature map in Fig. 4(c)]. The last CNN layer seen in Fig. 4(c) applies a kernel with the size of 3×3. This CNN layer accomplishes the feature extraction of spatial correlation and prediction, yielding the output expressed as $\boldsymbol{S}_{\text{spa}}$.

4) STFP Module: Finally, the STFP module, as shown in Fig. 4(a), is designed to incorporate the outputs $\boldsymbol{T}_{\text{tem}}$ by TAFE with the outputs $\boldsymbol{S}_{\text{spa}}$ of SAFE to mine the overall spatiotemporal correlation, and provide the Chl-a prediction for the whole lead time series. In detail, in the STFP module shown in Fig. 4(a), for each subregion relied model, both $\boldsymbol{T}_{\text{tem}}$ and $\boldsymbol{S}_{\text{spa}}$ are input to the "reshape&fusion" block to integrate the spatiotemporal features. In each of these blocks, $\boldsymbol{T}_{\text{tem}}$ with the size of $(C,(M\times N)/P)$ is first structured to have the same size as $\boldsymbol{S}_{\text{spa}}$, which is $(C,(M/(P)^{1/2}),(N/(P)^{1/2}))$, where C represents the number of channels. Then, the two feature maps are concatenated to form a new feature map of size $(2C,(M/(P)^{1/2}),(N/(P)^{1/2}))$, which is sent to the " predict" block.

The "predict" block is functioned to predict the future Chl-a values of each subregion via the association mining of the spatiotemporal fusion features. More specifically, the high-dimensional association mining of the spatiotemporal evolution features is achieved by the multiple cascaded CNN layers, which have the 3×3 sized convolution kernels. The prediction results with the size of $(L,(M/(P)^{1/2}),(N/(P)^{1/2}))$, namely, $\widehat{\boldsymbol{S}}_{(p,T+1:T+L)}$ shown in Fig. 4(a), are the outputs of the CNN layers. Finally, the Chl-a predictions of multiple subregions are joined in the "region concat" block to complete the future Chl-a prediction in the whole area considered. For clarity, the implementation procedures of our Ca-STANet for the Chl-a prediction are summarized in Algorithm 1.

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Algorithm 1 Ca-STANet Algorithm for Chl-a Prediction

Input: Historical Chl-a sequence $\boldsymbol{S}_{1:T}$, No. of sub-regions P, maximum epoch γ, learning rate λ, and prediction step size L.

Output: Future Chl-a sequence $\widehat{\boldsymbol{S}}_{T+1:T+L}$.

Step 1: ***Spatio-temporal input preprocessing (STIP)***

Construct the Chl-a sequence $\widetilde{\boldsymbol{S}}_{(p,1:T)}$, $p = 1, \cdots, P$ via sub-region division;

Construct the inputs TI_p and SI_p for each sub-region.

Step 2: ***Training Ca-STANet***

for $p = 1, \ldots, P$, **do**

  Initialize TAFE, SAFE and STFP modules;

  while *training epoch is less than* γ, **do**

    1) Generate temporal features $\boldsymbol{T}_{\mathrm{tem}}$ using TAFE module;

    2) Generate spatial features $\boldsymbol{S}_{\mathrm{spa}}$ using SAFE module;

    3) Generate spatio-temporal correlations by fusing $\boldsymbol{T}_{\mathrm{tem}}$ and $\boldsymbol{S}_{\mathrm{spa}}$ using STFP module;

    4) Predict $\widehat{\boldsymbol{S}}_{(p,T+1:T+L)}$ and update modules' parameters using gradient descent.

  end

end

Output $\widehat{\boldsymbol{S}}_{T+1:T+L}$ via concatenating $\widehat{\boldsymbol{S}}_{(p,T+1:T+L)}$, $p = 1, \cdots, P$.

IV. Experiments and Results

A. Gap Filling for Five-Day and Monthly Chl-a Data

Before gap filling, the parts with more than 90% of missing data in the five-day and monthly Chl-a data are filtered to ensure the accuracy of later reconstruction. Furthermore, in our study, 5% of the filtered data are randomly selected to form the cross-validation set so as to evaluate the performance of DINEOF. After cross validation, these 5% of data are then included to form the gap-free data for the following Chl-a prediction.

To evaluate the accuracy of the reconstructed Chl-a data, the cross-validation set of data after the DINEOF process is plotted against the original satellite-measured data, as shown in Fig. 7, for both the five-day and monthly Chl-a data. Here, the density of scatters is expressed by a kernel density estimation using the Gaussian kernel. The satellite-measured five-day and monthly Chl-a data vary in the range of 0.11–93.08 and 0.11–92.47 ($mg \cdot m^{-3}$), respectively. It can be seen from Fig. 7 that the data scatter around the 1:1 line in both Fig. 7(a) and (b), with the root-mean-square error (RMSE) of 0.88 and 1.25 ($mg \cdot m^{-3}$), respectively. Besides, the coefficient of determination (R^2) [52] is 0.89 for the five-day data and 0.83 for the monthly data. Furthermore, if we normalize the distribution using a natural logarithm transformation, the RMSE is 0.06 and 0.07, respectively, while R^2 is 0.93 for the five-day data and 0.92 for the monthly data. Based on these results, we can confident that DINEOF performs well in the reconstruction of the Chl-a data, which can be further exploited to analyze the spatial and temporal patterns in the Chl-a data of Bohai sea.

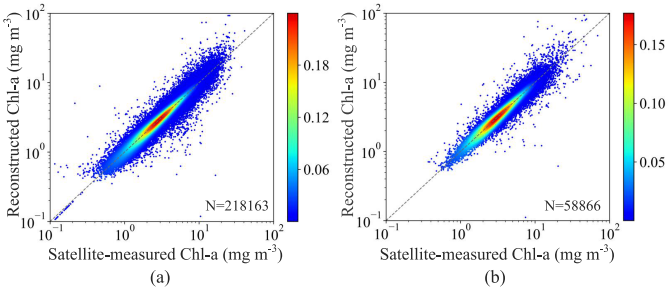


Fig. 7. Scatterplots of satellite-measured Chl-a data versus reconstructed Chl-a data. (a) Five-day Chl-a data. (b) Monthly Chl-a data.

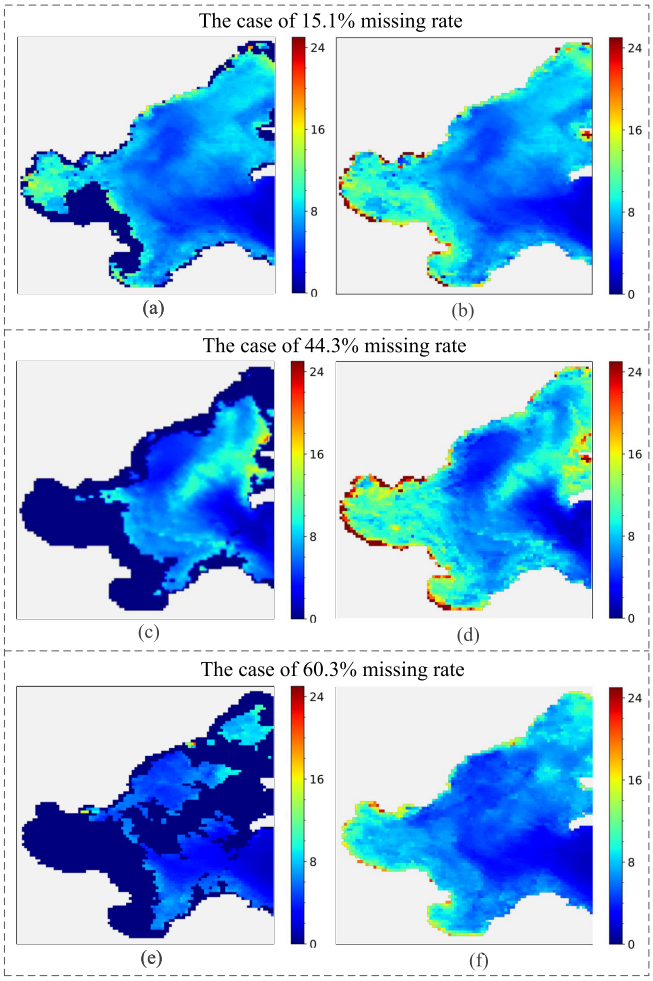


Fig. 8. (a) and (c) Original monthly data and five-day (e) Chl-a data, and their (b), (d), and (f) reconstructed data.

The original and reconstructed Chl-a data for three scenarios with different missing rates are shown in Fig. 8. The gray background represents the land or sea areas not involved in the reconstruction. The dark blue areas in Fig. 8(a), (c), and (e) are the missing parts needing reconstruction. The data missing rates for these scenarios are 15.1%, 44.3%, and 60.3%, respectively. On the right-hand side, Fig. 8(b), (d), and (f) gives the correspondingly reconstructed Chl-a data. From Fig. 8, it is clear that under various missing rates, the spatial distribution of Chl-a can be reconstructed with high accuracy.

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TABLE I
SUBREGION EXPERIMENT WITH MONTHLY DATA

Methods	Metrics	Prediction Months		
		1-step	3-step	6-step
whole-area	RMSE	1.91	2.39	2.41
	MAE	1.16	1.63	1.60
	PCorr	0.880	0.839	0.840
4 sub-regions	RMSE	1.83	2.15	2.17
	MAE	1.09	1.38	1.40
	PCorr	0.881	0.848	0.847
9 sub-regions	RMSE	1.82	2.12	2.11
	MAE	1.09	1.32	1.33
	PCorr	0.879	0.850	0.849
16 sub-regions	RMSE	1.85	2.17	2.13
	MAE	1.09	1.33	1.29
	PCorr	0.874	0.848	0.851

B. Experiment Settings for Prediction

Next, we show the performance of Chl-a prediction. Our forecasting algorithm is operated on a single NVIDIA 2080TI GPU. During training, the mean square error (MSE) as the loss function and the adaptive moment estimation (Adam) as the optimizer are applied. After reconstructing the corresponding Chl-a data, we select 70% of the data as the training samples, while the remaining 30% data are used as the testing data. Furthermore, the training and testing samples are chosen in the order of time rather than randomly, where the testing samples are the future samples relative to the training samples. The learning rate is set to 0.001 for the monthly data and 0.0001 for the five-day data. The size of the mini-batch is set to 8, and the model is trained by 40 epochs. Inspired by [39], which used the sea surface temperature-based spatiotemporal sequences of three consecutive months to predict the ENSO, our model uses the Chl-a data of three consecutive steps as the inputs and the direct strategy to perform the multistep-ahead forecasting. More specifically, for the five-day Chl-a data, the input data are constructed in the format: [1485, 3, 96, 96], where the first number is the number of the Chl-a samples, the next number is the length T of the historical Chl-a sequence, and the last two numbers are the height M and width N of the input data, respectively. Similarly, for the monthly Chl-a data, the input data are constructed in the format: [273, 3, 96, 96].

1) Evaluation Metrics: RMSE, mean absolute error (MAE) and Pearson correlation coefficient (PCorr) are used to assess the performance of the Ca-STANet, which are evaluated from the formulas of

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{n}(p_i - o_i)^2}{n}} \tag{6}$$

$$\text{MAE} = \frac{\sum_{i=1}^{n}|p_i - o_i|}{n} \tag{7}$$

$$\text{PCorr} = \frac{\sum_{i=1}^{n}(p_i - \bar{p})(o_i - \bar{o})}{\sqrt{\sum_{i=1}^{n}(p_i - \bar{p})^2 \sum(o_i - \bar{o})^2}} \tag{8}$$

where p_i is the predicted Chl-a value and o_i is the Chl-a value sensed by satellite, whose averages are denoted by $\bar{p}$ and $\bar{o}$, respectively, while n is the total number of testing samples.

2) Baseline Models: The following learning-based methods are selected as the baselines for comparison with our model, which are as follows.

1) *CNN [53]:* Note that the CNN model is slightly modified by removing the fully connected layer and a max-pooling layer, when the Chl-a in large-scale area is predicted in our study.
2) *LSTM-S [54]:* The LSTM-S model uses all the points in a field as one sample. However, in our study of simultaneously considering a large number of points, the model easily leads to memory overflow. To solve this problem, the original area is divided into two subareas for training.
3) *CNN-LSTM [27]:* The CNN-LSTM, consisting of three convolutional layers, a flattening layer, and an LSTM layer, was proposed to predict two water quality variables, i.e., dissolved oxygen and Chl-a.
4) *ConvLSTM [43]:* ConvLSTM was proposed to overcome the drawbacks of LSTM in handling the spatiotemporal data. The ConvLSTM-assisted Chl-a prediction model is the same as that in [43].

C. Setting of Parameter P

In this section, the influence of the number of subregions P on the prediction performance is studied. It can be understood that using the whole-area sequences as inputs can also achieve good performance for the large-area Chl-a prediction, if there are enough training samples available. However, when only thousands of five-day or monthly data are available for Chl-a prediction, the performance achieved is generally limited, which can be improved by dividing the whole region into subregions. Table I shows the performance results of the monthly prediction, when different numbers of subregions are employed. It can be seen that the performance of using 4, 9, or 16 subregions is superior to that of using the whole area, i.e., one region. Furthermore, it is observed that there exists an appropriate number of subregions that results in the best performance. As seen in Table I, for the cases considered, this is given by the case with nine subregions. In this case, the Ca-STANet is capable of making a relatively clear distinction between the subregions in the shallow waters with high fluctuation and those far away from land with a low variation. By contrast, if there are too many tiny subregions, some of the neighborhood spatial information may be missed by the Ca-STANet, yielding the degraded prediction accuracy. Furthermore, a bigger number of subregions results in the expansion of model parameters, which, in turn, results in bigger complexity for the model training. Therefore, in our following studies, we set P to 9.

D. Experimental Results and Analysis

Now, we demonstrate and compare the Chl-a prediction results by different methods. During the experiments, all the baseline models are trained using the same hyperparameters and prediction strategy, as mentioned during the Ca-STANet training in Section IV-B.

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TABLE II
PREDICTION RESULTS ON THE CHL-A DATASETS OF BOHAI SEA

Models	Metrics	Prediction 5-days				Prediction Months		
		1-step	3-step	6-step	9-step	1-step	3-step	6-step
CNN	RMSE	1.86	1.96	2.11	2.18	2.31	2.69	2.84
	MAE	1.33	1.41	1.56	1.64	1.52	1.81	1.90
	PCorr	0.848	0.822	0.798	0.787	0.870	0.799	0.770
LSTM-S	RMSE	1.85	1.89	2.00	2.06	2.67	2.62	2.59
	MAE	1.22	1.24	1.32	1.38	1.74	1.68	1.67
	PCorr	0.828	0.826	0.820	0.809	0.829	0.812	0.814
CNN-LSTM	RMSE	1.74	1.74	1.80	2.02	2.11	2.47	2.44
	MAE	1.25	1.26	1.31	1.58	1.41	1.72	1.69
	PCorr	0.836	0.826	0.824	0.809	0.856	0.827	0.827
ConvLSTM	RMSE	**1.46**	1.62	1.83	2.01	1.90	2.55	2.76
	MAE	**0.98**	1.12	1.33	1.52	1.14	1.72	1.91
	PCorr	0.857	0.833	0.807	0.787	0.870	0.795	0.775
Ca-STANet	RMSE	1.47	**1.58**	**1.73**	**1.85**	**1.82**	**2.12**	**2.11**
	MAE	1.00	**1.11**	**1.27**	**1.39**	**1.09**	**1.32**	**1.33**
	PCorr	**0.866**	**0.849**	**0.834**	**0.826**	**0.879**	**0.850**	**0.849**

Table II shows the RMSE, MAE, and PCorr values obtained by the different prediction algorithms. From Table II, we can see that for the various steps of predictions based on the five-day data and monthly data, CNN achieves the poorest performance, as it does not exploit the chronological correlation of the input data. LSTM-S performs slightly better than CNN, especially for the long-term prediction. This is because LSTM-S has the ability to mine the long-term dependency of the Chl-a data by its recurrent network structure and gating mechanisms. CNN-LSTM and ConvLSTM, which make use of the spatiotemporal correlation of input data, outperform both the CNN, which only mines the spatial correlation, and the LSTM-S, which focuses only on the time dependency. As shown in Table II, for the relatively short-term prediction, ConvLSTM outperforms CNN-LSTM. In particular, for the one-step prediction, ConvLSTM attains the best RMSE and MAE performance among the five algorithms considered. By contrast, CNN-LSTM may be superior to ConvLSTM in medium and relatively long-term prediction, especially in the monthly data case. The possible reason behind is that the structure of ConvLSTM is relatively complex, making it prone to the inadequate training for the small sample size, resulting in unstable performance for multistep prediction.

From Table II, we can explicitly see that our proposed model outperforms all the other considered models and attains the best performance in nearly all the cases. Specifically, for the monthly data, it outperforms the ConvLSTM by 0.08, 0.43, and 0.65 in terms of RMSE, when the one-step, three-step, and six-step predictions are implemented. In terms of MAE, our method outperforms ConvLSTM by 0.05, 0.40, and 0.58, also when these monthly predictions are considered. In terms of the PCorr, the result achieved by Ca-STANet is higher by 0.009, 0.055, and 0.074 than that obtained by ConvLSTM for the one-step, three-step, and six-step predictions. For the five-day data, our Ca-STANet outperforms all the other considered models in terms of RMSE and MAE performance and has the capability to mine the correlation existing in the data.

In general, Table II shows that the prediction error of all the five models becomes larger when the step size of prediction becomes bigger, but among them, our model increases at the slowest pace. This implies that introducing the TAFF module enables Ca-STANet to efficiently extract the feature of the long-term dependency that exists in the Chl-a data sequences. As shown in Table II, our Ca-STANet significantly outperforms ConvLSTM for the monthly prediction relying on the small size data samples. The reason behind is that, by dividing the data of Bohai Sea into subregions according to the spatial heterogeneity, our method is capable of fully exploiting the characteristics of each subregion in the case of a relatively small number of samples. On the other side, our method can efficiently fuse the regional features and global features to provide the reliable and robust long-term prediction. Owing to the abovementioned experimental results, our approach is, in general, capable of outperforming the legacy schemes for the Chl-a prediction.

Figs. 9 and 10 depict the RMSE spatial distributions obtained by the Ca-STANet for the five-day and monthly data, respectively, which were calculated grid-by-grid from the prediction errors of all the samples during the testing period. Explicitly, the RMSE in the coastal waters is higher than that in the other offshore waters for both datasets. This is because in the coastal waters, there are large Chl-a spatial gradient, fast temporal variation, and outbreak of red tides. From Fig. 9, we observe that as the step size of prediction increases, the RMSE increases quickly at the locations where Chl-a changes fast on the five-day basis. By contrast, the RMSE is relatively stable at the locations in the deep offshore waters in the Bohai Sea. For instance, the RMSE values are 1.41, 1.55, 1.84, and 2.09, respectively, when one-step, three-step, six-step, and nine-step predictions are executed within area A (bounded by 38.50°–39.00°N, 118.50°–119.13°E, in Fig. 9) that is close to the shoreside. While for area B (bounded by 38.00°–38.63°N, 121.00°–121.50°E, in Fig. 9) in the deep offshore waters, the corresponding RMSE values are 0.47, 0.54, 0.61, and 0.67, respectively. Furthermore, it is shown that for the one-step to nine-step prediction, the proportion of the grid points with their RMSE below 2 ($mg \cdot m^{-3}$) is 92.4%, 89.9%, 83.2%, and 75.0%, respectively. A similar comparison in the context of the monthly Chl-a data can be obtained based on Fig. 10. From these results, it can be implied that the proposed model

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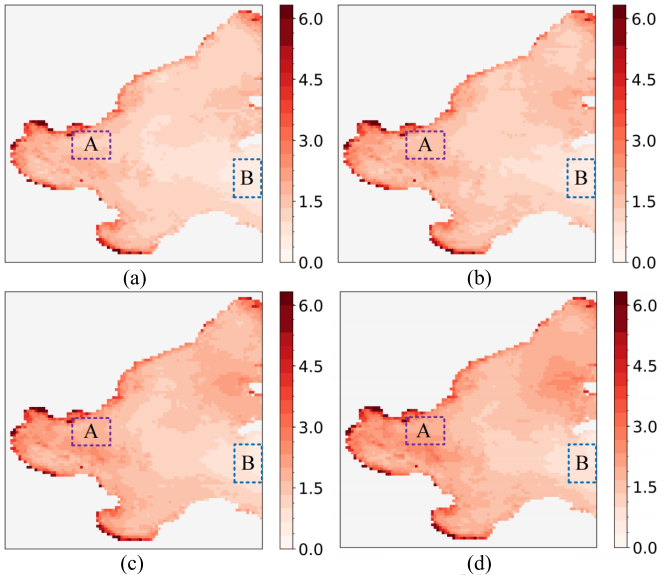


Fig. 9. Spatial distributions of RMSE calculated from all testing samples of the five-day Chl-a data, where one prediction step represents a five-day interval. Specifically, area A is close to the shore, and area B is far away from the shore and in deep offshore waters. (a) One-step prediction. (b) Three-step prediction. (c) Six-step prediction. (d) Nine-step prediction.

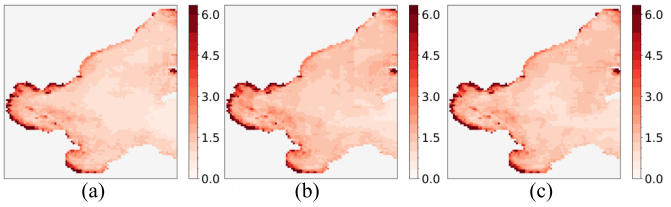


Fig. 10. Spatial distributions of RMSE calculated from all testing samples of the monthly Chl-a data, where one prediction step represents one month. (a) One-step prediction. (b) Three-step prediction. (c) Six-step prediction.

enables a stable prediction performance for most of the areas in the Bohai Sea.

The temporal trends of the real and predicted Chl-a values for the five-day and monthly data are present in Figs. 11 and 12, respectively, which were calculated sample-by-sample from the forecasting results. Specifically, Fig. 11 compares the predicted Chl-a values with the real values on the five-day intervals from 2018 to 2020. It can be observed that the predicted results by the Ca-STANet fit well with the real Chl-a data. Apparently, as the number of prediction steps increases, the prediction accuracy decreases. Similarly, Fig. 12 compares the predicted Chl-a values with the real Chl-a values on the monthly data basis between January 2014 and December 2020. As shown by the results, the real Chl-a exhibits clearly the monthly periodic characteristics. Our proposed method can efficiently extract this periodicity during the training procedures, resulting in a good periodic fitting with the real Chl-a data for all the prediction steps considered. Furthermore, with the increase in the prediction step size from one to six months, error does not increase significantly. The reason behind this is that the monthly data have a seasonal cycle, and our proposed method can improve the accuracy of long-term prediction. In general, the results of Figs. 11 and 12 explain that the proposed Ca-STANet is

TABLE III
ABLATION EXPERIMENT WITH FIVE-DAY DATA

Method	RMSE(mg · m^{-3})			
	1-step	3-step	6-step	9-step
Model_base	1.86	1.96	2.11	2.18
Model_S	1.69	1.83	1.98	2.10
Model_ST(w/oAtt)	1.46	1.58	1.74	1.87
Model_ST	1.47	1.58	1.73	1.85
Method	MAE(mg · m^{-3})			
	1-step	3-step	6-step	9-step
Model_base	1.33	1.41	1.56	1.64
Model_S	1.20	1.34	1.49	1.61
Model_ST (w/oAtt)	0.99	1.11	1.28	1.41
Model_ST	1.00	1.11	1.27	1.39
Method	PCorr			
	1-step	3-step	6-step	9-step
Model_base	0.848	0.822	0.798	0.787
Model_S	0.860	0.833	0.812	0.801
Model_ST (w/oAtt)	0.866	0.850	0.833	0.825
Model_ST	0.866	0.849	0.834	0.826

capable of efficiently making use of the temporal correlation for improving the accuracy of Chl-a prediction. However, due to the complexity and chaos of the climate variability, the proposed model underestimates the Chl-a peaks in the winter but overestimates the Chl-a valleys in the summer, especially when the number of prediction steps increases.

E. Ablation Study

Next, we carry out an extensive ablation study to demonstrate the effectiveness of the SAFE module and the TAFE module in our Ca-STANet. To serve the purpose, four models are considered, including the following: 1) *Model_base*, which only uses three layers of the CNN with the original inputs instead of the outputs of the STIP module; 2) *Model_S*, which includes only the SAFE module; 3) *Model_ST(w/oAtt)*, which includes the SAFE module and the TAFE module but without the multihead attention layer; and 4) *Model_ST*, which includes both the SAFE module and the TAFE module with the multihead attention layer.

Experimental results are shown in Tables III and IV for the five-day and monthly data, respectively, where one step represents a five-day interval in Table III but one month in Table IV. As shown in the tables, by using the Model_S, the RMSE and MAE for both the five-day and monthly data are reduced in comparison with that of the Model_base. Specifically, the MAE of Chl-a prediction is decreased by 0.2, 0.36, and 0.53, when one-step, three-step, and six-step predictions are, respectively, carried out on the monthly data. The Model_S can also improve the PCorr effect. As shown in Table III, PCorr is increased by 0.006, 0.016, 0.022, and 0.025, respectively, when the one-step, three-step, six-step, and nine-step Chl-a predictions are applied. The results demonstrate that the SAFE module is capable of mining the spatial correlations existing in both the subregions and the global region, yielding an improved prediction performance.

From Tables III and IV, we can explicitly see that Model_ST (w/oAtt) outperforms Model_S in both the five-day and monthly Chl-a prediction. This implies that the introduction

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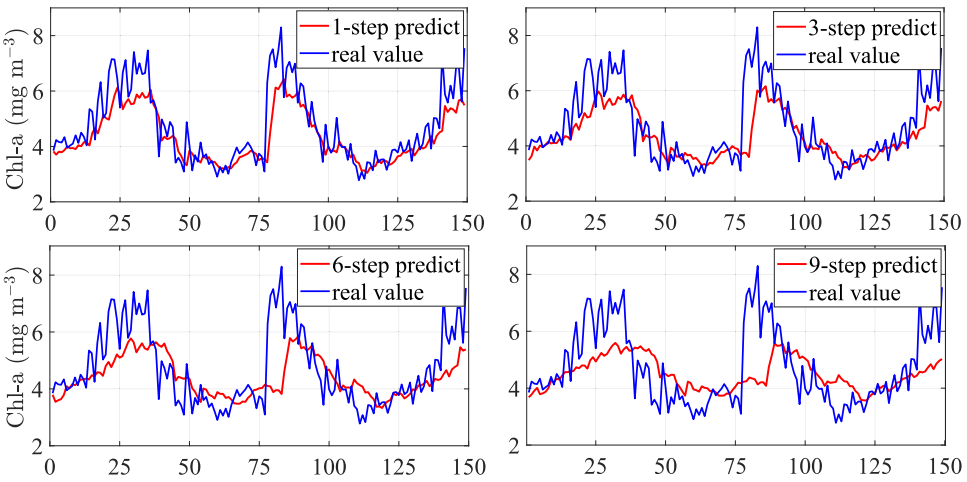


Fig. 11. Temporal trend of the real and predicted Chl-a values for the different step sizes, when the prediction step size is five days.

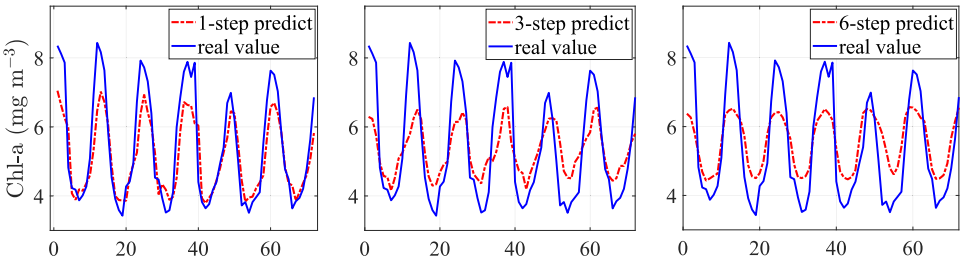


Fig. 12. Temporal trend of the real and predicted Chl-a values for different step sizes, when the prediction step size is one month.

TABLE IV
ABLATION EXPERIMENT WITH MONTHLY DATA

Method	RMSE(mg · m^{-3})		
	1-step	3-step	6-step
Model_base	2.31	2.69	2.84
Model_S	2.05	2.25	2.22
Model_ST (w/oAtt)	1.89	2.22	2.23
Model_ST	1.82	2.12	2.11
Method	**MAE(mg · m^{-3})**		
	1-step	3-step	6-step
Model_base	1.52	1.81	1.90
Model_S	1.32	1.44	1.37
Model_ST (w/oAtt)	1.11	1.41	1.34
Model_ST	1.09	1.33	1.30
Method	**PCorr**		
	1-step	3-step	6-step
Model_base	0.870	0.799	0.770
Model_S	0.878	0.840	0.839
Model_ST (w/oAtt)	0.879	0.846	0.843
Model_ST	0.879	0.850	0.849

TABLE V
INFLUENCE OF WINTER GAP-FILLED DATA ON PREDICTION IN SPRING, SUMMER, AND AUTUMN

Data	Metrics	Prediction Months		
		1-step	3-step	6-step
All-data	RMSE	1.68	1.94	2.08
	MAE	0.95	1.20	1.31
Except-winter	RMSE	1.66	1.82	1.96
	MAE	0.94	1.14	1.24

of a time branch for extracting the temporal features is important for improving the prediction accuracy. Furthermore, Model_ST, which adds an extra multihead attention layer on Model_ST (w/oAtt), performs better than Model_ST (w/oAtt) in most cases. This advantage is obtained, because the added attention layer can implement the differential modeling for each output vector based on its different correlation attention scores within all inputs, which enhances the mining of the long-term dependencies of the entire historical Chl-a inputs.

Consequently, it can be seen from Tables III and IV that the Model_ST, in general, achieves the best RMSE and MAE performance and yields the highest PCorr values. Therefore, by introducing the TAFE and SAFE modules to the Ca-STANet, the prediction performance can be improved, especially for the long-term prediction, owing to that it can simultaneously extract the temporal and spatial features for the fusion prediction. This implies that the correlations in both the temporal domain and spatial domain should be exploited to increase the prediction accuracy.

V. ANALYSIS OF RESULTS AND INFLUENCE OF WINTER DATA

Sea ice and heavy cloud cause a high loss of Chl-a data of Bohai Sea in winter. Hence, the winter Chl-a data, especially in the ice-covered area, are mainly gap-filled data obtained by the DINEOF method. Therefore, it is important to investigate the impact of the winter gap-filled data on the Chl-a prediction in spring, summer, and autumn, as well as on the accuracy of Chl-a prediction in winter. To serve this purpose, we set up the following experiments. First, we only consider the Chl-a prediction in spring, summer, and autumn to analyze the impact of the winter gaped-filled data on the performance

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TABLE VI
PREDICTION RESULTS FOR SOME GRID POINTS WITH TRUE CHL-A VALUES IN WINTER

Grid	Dataset	Miss_rate	Steps	True_v	P_gap	P_w/ogap
G1	monthly	0.4%	1-step	5.37	4.90(Δ-0.45)	4.85(Δ-0.52)
G2	monthly	23.9%	3-step	11.18	10.17(Δ-1.01)	6.04(Δ-5.14)
G3	monthly	55.1%	1-step	7.48	9.84(Δ+2.36)	0.86(Δ-6.62)
G4	5-day	23.8%	1-step	3.37	3.29(Δ-0.08)	3.57(Δ+0.20)
G5	5-day	33.8%	3-step	1.95	1.67(Δ-0.28)	0.98(Δ-0.97)
G6	5-day	79.7%	1-step	7.75	8.90(Δ+1.15)	1.77(Δ-5.98)

of the Ca-STANet. As shown in Table V, "all data" means that all four seasons' data are used for training, whereas "except winter" means that only the data in spring, summer, and autumn seasons are used for training. Table V explicitly shows that the corresponding results obtained by including and without including the training data of winter season agree well with each other and have only small differences for the monthly Chl-a prediction in spring, summer, and autumn.

Second, to investigate the prediction accuracy of winter season data, some grid points with the true satellite-measured Chl-a values at different times are chosen to verify the performance of the Ca-STANet. The settings and results are shown in Table VI, "Miss_rate" represents the miss rate of the grid points in the monthly or five-day Chl-a dataset, "True_v" represents the true value of the grid point at a certain time, "P_gap" represents the predicted Chl-a value for the certain time by using the gap-filled data as inputs, "P_w/ogap" represents the predicted Chl-a value for the certain time by using the original incomplete data as inputs, and, finally, Δ represents the difference between the predicted value and the true Chl-a value. From Table VI, we can clearly see that the differences between "P_gap" and "True_v" are relatively small for the six grid points with different miss rates and Chl-a concentrations. By contrast, the differences between "P_w/ogap" and "True_v" are typically several times bigger than the corresponding differences between "P_gap" and "True_v." This is especially the case, when the "Miss_rate" is high. Hence, even though the DINEOF algorithm fills the missing data of the Bohai Sea in the winter without considering in detail the ice-covered areas, the prediction results obtained by Ca-STANet based on the gap-filled data have much higher accuracy than that obtained based on the original incomplete data. Therefore, we can conclude that the gap-filled data by the DINEOF algorithm are effective for Ca-STANet to achieve good performance.

VI. CONCLUSION

Accurate Chl-a prediction is paramount for providing an early warning of red tide and keeping the marine ecosystem healthy. To the best of our knowledge, most existing learning-based Chl-a prediction approaches predict the Chl-a of a single point or multiple points in a small region, based mainly on the monitoring data. By contrast, the long-term Chl-a prediction in the relatively large-scale area is still in its infancy. In this article, from the perspective of the spatiotemporal field, the remote sensing data of OC-CCI were used to predict the Chl-a in the whole Bohai Sea. First, the DINEOF method was introduced to fill the gap in the OC-CCI Chl-a data between 1980 and 2020. For prediction, a new framework, namely, the Ca-STANet, was designed to predict the Chl-a in a large-scale area by exploiting both the spatial and temporal correlations. In our method, the original input data were divided into multiple subregions not only to tackle the spatial heterogeneity of large-scale areas but also to avoid the occurrence of catastrophic forgetting. A TAFE module was designed to derive the important time features and capture the long-term dependency. An SAFE module was proposed to extract the spatial evolution features in the subregions and, at the same time, mine the dynamic correlation of global spatial neighborhood. Owing to these well-designed modules, Ca-STANet is capable of capturing the spatiotemporal correlations to achieve highly reliable Chl-a prediction. The experimental results demonstrated that the Ca-STANet achieves better performance than the state-of-the-art methods in both the five-day and monthly Chl-a predictions. Moreover, Ca-STANet can even perform efficiently with a small number of data samples, such as the monthly Chl-a data.

In this article, the missed Chl-a data in the Bohai Sea are reconstructed by the DINEOF without considering in detail sea-ice coverage. In the future, we will make an endeavor to study the high-quality Chl-a dataset reconstruction by considering the dynamic sea-ice mask and investigate its impact on our Ca-STANet and other Chl-a prediction algorithms. Moreover, in this article, the introduced Chl-a prediction depends only on Chl-a data. In practice, however, Chl-a may be affected by the other hydrological and meteorological conditions, as well as the land-sourced pollutant emission. Therefore, in our future work, we will endeavor to further improve our Ca-STANet by exploiting the influence of these factors.

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Min Ye is currently pursuing the Ph.D. degree with the School of Information Science and Engineering, Ocean University of China, Qingdao, China.

From 2021 to 2022, she was a Visiting Ph.D. Student with the School of Electronics and Computer Science, University of Southampton, Southampton, U.K. Her major research interests include remote sensing data processing, marine environmental forecasting, and machine learning.



Bohan Li received the Ph.D. degree from the University of Southampton, Southampton, U.K., in 2022.

He is currently a Research Fellow with the University of Leicester, Leicester, U.K. In recent years, his research has been mainly focused on the multicarrier-division duplex-based wireless communications. His research interests include the various techniques for the future generations of wireless systems, including full duplex communications, mmWave massive MIMO, cell-free massive MIMO, deep-reinforcement learning, and the graph neural networks for optimization of wireless systems.



Jie Nie (Member, IEEE) received the Ph.D. degree in computer science from the Ocean University of China, Qingdao, China, in 2011.

From September 2009 to September 2010, she was a Visiting Scholar with the School of Electrical Engineering, University of Pittsburgh, Pittsburgh, PA, USA. From 2015 to 2017, she was a Post-Doctoral Researcher with Tsinghua University, Beijing, China. She is currently working with the Ocean University of China. Her research interests include mainly social media and multimedia content analysis.



Yuntao Qian (Senior Member, IEEE) received the B.E. and M.E. degrees in automatic control from Xi'an Jiaotong University, Xi'an, China, in 1989 and 1992, respectively, and the Ph.D. degree in signal processing from Xidian University, Xi'an, China, in 1996.

From 1996 to 1998, he was a Post-Doctoral Fellow with Northwestern Polytechnical University, Xi'an, China. Since 1998, he has been with the College of Computer Science, Zhejiang University, Hangzhou, China, where he became a Professor in 2002. He was a Visiting Professor with Concordia University, Canada, from 1999 to 2001; Hong Kong Baptist University, Hong Kong, in 2002; Carnegie Mellon University, Pittsburgh, PA, USA, in 2006; the Canberra Research Laboratory of NICTA, Australia, in 2010; Macau University, Macau, China, from 2015 to 2016; and Griffith University, Nathan, QLD, Australia, in 2018. His research interests include machine learning, signal and image processing, pattern recognition, and hyperspectral imaging.

Dr. Qian is currently an Associate Editor of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING.



Lie-Liang Yang (Fellow, IEEE) received the B.Eng. degree in communications engineering from Shanghai TieDao University, Shanghai, China, in 1988, and the M.Eng. and Ph.D. degrees in communications and electronics from Northern (Beijing) Jiaotong University, Beijing, China, in 1991 and 1997, respectively.

He is currently a Professor of Wireless Communications with the School of Electronics and Computer Science, University of Southampton, Southampton, U.K. He has published more than 400 research papers in journals and conference proceedings, authored/coauthored three books, and also published several book chapters. The details about his research publications can be found at https://www.ecs.soton.ac.uk/people/llyang. His research interests include wireless communications, wireless networks and signal processing for wireless communications, and molecular communications and nanonetworks.

Prof. Yang is a fellow of the IET. He was a Distinguished Lecturer of the IEEE VTS. He served as an Associate Editor for various journals and is currently a Senior Editor of the IEEE ACCESS and a Subject Editor of the *Electronics Letters*. He also acted different roles for organization of conferences.

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